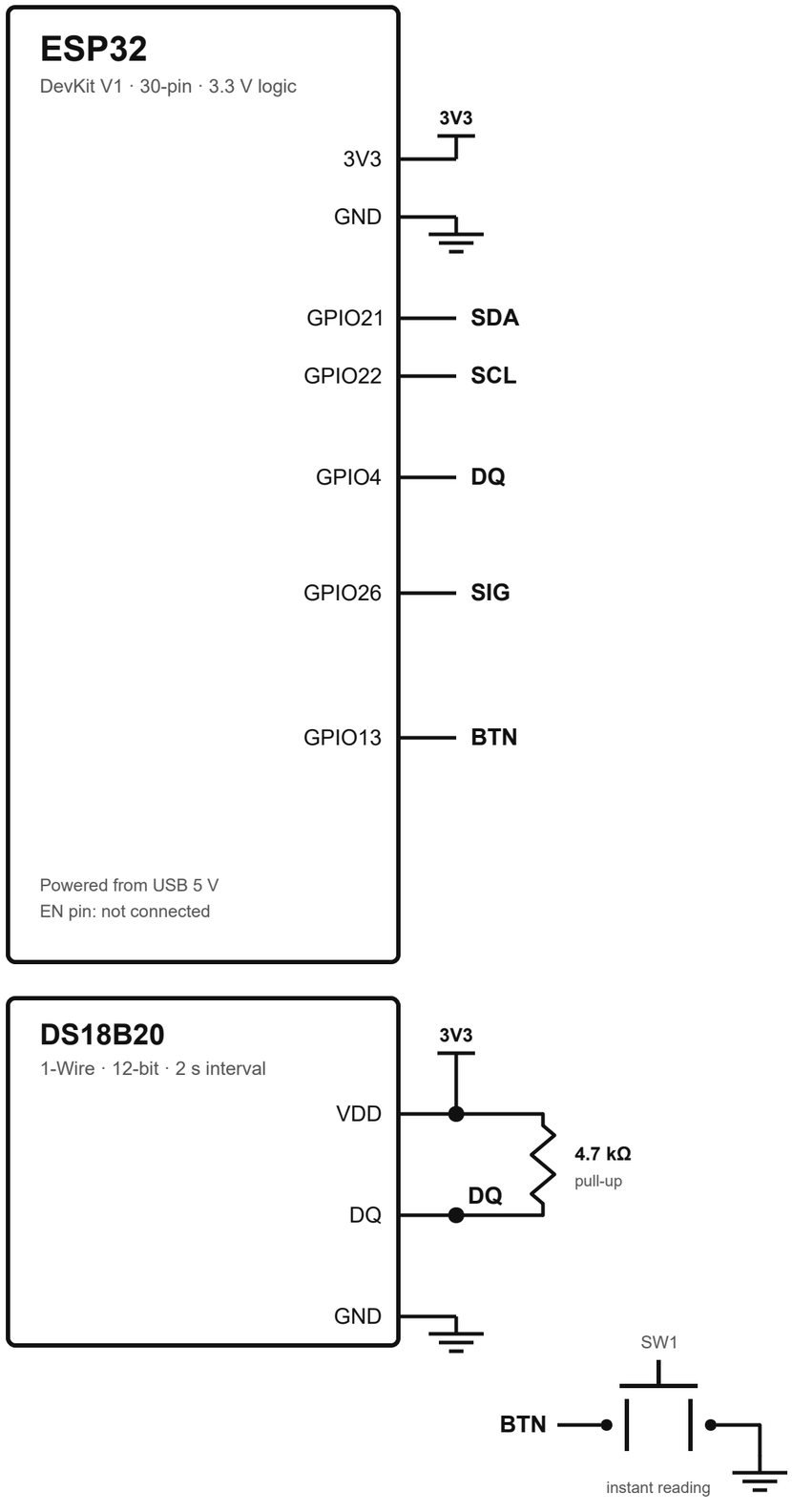
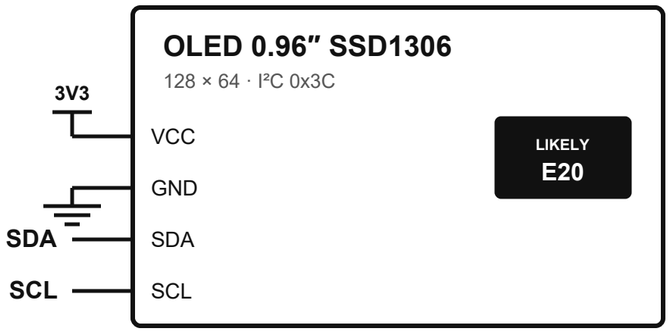
ETHANOLBLEND

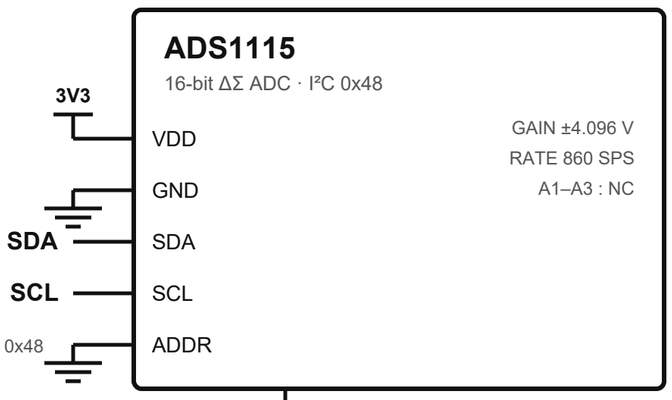
Measurement schematic — ESP32 + ADS1115 conductivity probe

20 Hz square excitation · single-ended ADC · 6-point empirical calibration

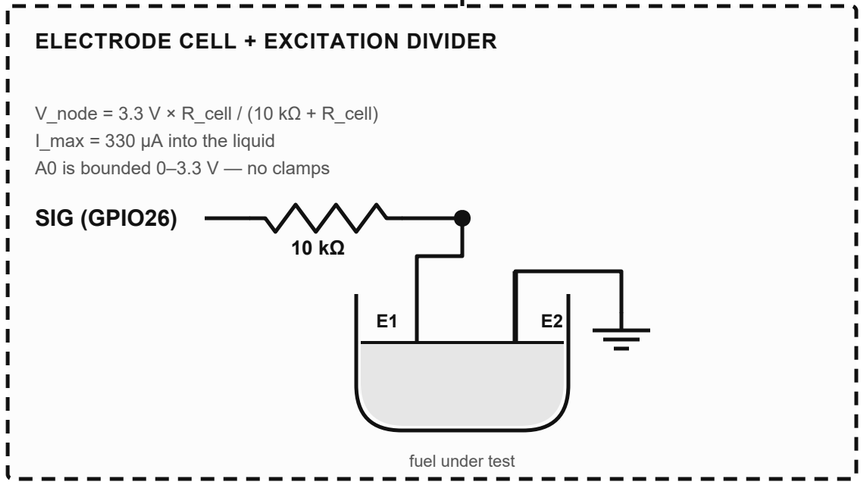
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DESIGN NOTES

Excitation: 20 Hz, 0–3.3 V,
50 % duty, GPIO26 hardware timer.

Node voltage
V_node = 3.3 × R_cell / (10 kΩ + R_cell)

Max current into the liquid:
3.3 V / 10 kΩ = 330 µA.

A0 is bounded 0–3.3 V by
construction — no clamps fitted.

ADC: single-ended, ±4.096 V,
860 SPS, 48-sample burst every 5 s.
Lookup key = max of the burst.

All logic 3.3 V. Never apply 5 V
to ADS1115 inputs (abs max 3.6 V).

Not fitted: HX711, load cell,
ICL8038, LM358.

Nets with the same name are joined.

SYMBOL LEGEND

- connection (nets joined)
- common ground (0 V)
- 3V3 rail
- no connect
- SDA — net label

Same net name = same electrical node.